Integrated Crowd Counting System Utilizing IoT Sensors, OpenCV and YOLO Models for Accurate People Density Estimation in Real-Time Environments

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Abstract- **Real-time crowd monitoring plays a pivotal role in effectively managing public spaces and ensuring safety. This study investigates the fusion of IoT devices and the YOLO object detection model to accurately count crowds. IoT devices facilitate the instantaneous collection of data from cameras, while the YOLO model adeptly identifies individuals within recorded video frames. The study rigorously assesses the performance of three YOLO variants: YOLO V5, YOLO V8 and YOLO V8 NAS. Findings reveal that YOLO V8 NAS surpasses YOLO V5 and YOLO V8 in mean average precision (mAP), achieving an exceptional mAP of 95.1%. This heightened precision is attributed to the integration of Neural Architecture Search (NAS) into the YOLO V8 NAS model, fine-tuning its architecture specifically for crowd counting tasks. It analyzes various networking models proposed in earlier studies for analyzing crowded scenes in public spaces. It emphasizes the potential of a hybrid model involving an IP camera module and a Deep Neural Network for effective crowd sensing. In this setup, the IP camera captures video footage, while the DNN detects individuals and assesses crowd density based on the count of people recognized. This approach presents an encouraging solution for real-time crowd monitoring and management in public environments.**

Keywords: MCS, IOT, OpenCV, YOLO, Face Detection, Transfer Learning, ILR.

I. INTRODUCTION

In last decade, there was a tremendous growth in sensor technology. Sensors are used to sense different physical phenomenon of the environment. Crowed sensing plays a crucial role in our everyday life like in Road transportation, Smart City, and Health care management system etc. Crowd sensing, also known as mobile crowd sensing, A large number of people with sensing mobile devices work together to share data and extract information for processes that are of mutual interest, such as measuring, mapping, analysing, and estimating.Mobile and wireless communication technologies are in high demand, efficient and high capability networks allow billions of connectivity simultaneously. Now, Mobile communication is used in a wide verity of application because of its high computational power technology (4G/5G etc.) and reliability. Various sensing devices are now integrated with Mobile and wireless communication for better sensitivity. These are making a bridge in between various physical world objects and generate information about environment themselves. Broadly this sensing is differentiated into two parts personal and community sensing. Personal sensing is interested on individual movement pattern monitoring whereas community sensing (participatory sensing) is for large scale monitoring. Latter one sense a wide spectrum of user involvement, hence it emerges a new term Mobile crowed sensing.

MCS is cloud-based and allows for the exchange of information between users as well as the use of a large number of smartphones for activities that have a significant public impact. Information gathered from smartphones with the aid of the cloud can be combined using data fusion techniques. Therefore, mobile sensing can be used as a flexible platform to support a variety of applications, from safety and smart city applications to environmental monitoring. It can also replace static sensing infrastructure.

Affordable and widely accessible, mobile sensing devices offer an abundance of sensing data. Today's smartphones typically contain a number of sensors for different uses. There are 59.6 million iPhone users worldwide according to one database, and their devices have built-in sensors like GPS, accelerometer, gyroscope, ambient light, proximity, microphone, and camera sensors. Light sensors are used to fine-tune the screen brightness. The phone can use both light and proximity sensors to perform basic context recognition tasks related to the user interface. Mobile social networking apps, navigation and location-aware searches are all made possible by the GPS.This study report outlines its development, application, and assessment. We present a novel solution as fig.1 to crowd count problems by combining the power of OpenCV and the YOLO model which enhance the administrative efficacy.

Fig 1. The estimated size of head in dense crowd scene (the bounding box represents the estimated head size)

Our proposed model explains the ideas of mobile crowd sensing and how it can be used in everyday life. The design and execution of mobile crowd sensing is described in fig. 2. Once the data dependability challenges are effectively handled, mobile sensing will become an investigated approach for obtaining sensing data from the physical world. We presented ILR, a location reliability protocol, as a first step toward establishing data reliability in sensed data (Improving Location Reliability). ILR also detects erroneous location claims in the sensed data and used YOLO model detect the person and count accurately.

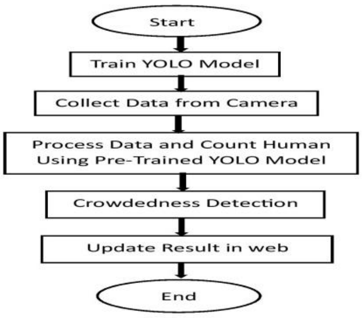

Fig 2. Proposed Model flow work

II. LITERATURE REVIEW

Nagoriya et al [1] stress the need for an innovative attendance management system using face recognition technology via Raspberry Pi. The system efficiently detects and identifies individuals, utilizing OpenCV and Eigenfaces algorithm for accuracy. Contrasting traditional methods, it aims for a non-intrusive, reliable, and cost-effective solution for attendance management in academic settings.Chukwude M. C. et al. [2] introduced "Roll Call," a web-based attendance system using Python, OpenCV, and Sci-kit Learn for face recognition at the University of Ilorin's Faculty of Engineering. Addressing flaws in traditional methods, Roll Call offers efficient attendance tracking via biometric face recognition, simplifying record-keeping with a user-friendly interface for both lecturers and students.Khaled E. et al. [3] introduce an automated exam attendance system based on face recognition. Employing LBP for facial recognition and Haar-Cascade/MTCNN for face detection, the system demonstrates robustness in varied conditions. Implemented in Python, it leverages AI algorithms and open-source libraries, presenting a reliable and cost-effective alternative to conventional attendance methods.Kristina R. et al. [4] explore using UAVs and deep learning for deer detection in Serbian forests. Various CNN architectures, including YOLO and SSD, are compared based on mAP, precision, and recall. YOLO achieves high precision (86%) and recall (75%), detecting deer accurately. The study highlights the potential for wildlife monitoring and underscores CNNs' efficacy in animal detection using UAVs.Ashwin R. et al. [5] present "AttenFace," a face recognition-based attendance system for educational institutions. Utilizing deep learning models, it automates attendance tracking, offering students flexibility while addressing accuracy challenges in varying conditions. While promising, improvements are needed for heightened accuracy and reliability in challenging scenarios. Sailendra K et al. [6] introduce a CNN-based facial recognition system, emphasizing its broad applications in security, biometrics, and surveillance. Their approach, leveraging deep learning, showcases advancements surpassing previous methods. The CNN model enables real-time identification, reducing processing power and time while maintaining high accuracy through feature selection, extraction, and training.ZareiNet al. [7] present Fast-Yolo-Rec, a method combining YOLO-based detection and LSTM-based position prediction networks for real-time vehicle detection. Integrating a spatial semantic attention mechanism (SSAM), it enhances accuracy and speed by alternating detection and prediction cycles. Evaluation on a Highway dataset shows its superior performance, offering a promising balance between speed and accuracy in vehicle detection.Suhane. A K et al. [8] explore YOLO's applicability in human detection and crowd counting, highlighting its significance in surveillance, security, and traffic analysis. While acknowledging YOLO's efficiency, they address challenges in complex scenes and object detection limitations, proposing pre/post-processing steps and emphasizing context-based utilization for optimal performance.Song. L et al. [9] introduce YOLO-T for vehicle logo detection (VLD), employing spatial structural correlation and multi-scale detection tailored for this task. They utilize position correlation between vehicle lights and logos, reducing background interference in the region of interest. Experimental validation on LOGO-17 dataset demonstrates superior VLD performance, mitigating environmental interference and enhancing accuracy. Swain S. et al [10]proposed a model on IoT and machine learning for heart disease diagnosis and prevention. He explores IoT device advancements, various ML approaches such as Decision Trees, and proposes combined models for precise diagnosis, highlighting their potential to enhance heart healthcare.Jayasing S. K. et al [11] introduce fire detection advancements, focusing on OPCNN's superior accuracy (95.11%) compared to CNN and J48 models, author collect data for this research using IoT devices in real time. Patra K.

J. et al [12]introduces a real-time attendance system merging OpenCV and YOLO, spotlighting YOLO V8 NAS for enhanced precision. It addresses research gaps with comprehensive methodology, reaching a 92.4% amAP and handling class imbalance through dataset measures.

III. RESEARCH DESIGN

A. Iot Framework for Mobile Crowd Sensing in Smart Cities Using a Single M2M Architecture

It also shows how MCS scenarios to address smart city issues can be enabled by an IOT framework. The advantages and disadvantages of the available options were shown by our literature analysis. This work contributes significantly by offering a framework for developing crowd sensing mobile applications that is both context and power aware. It also offers a data-driven method for cooperative crowd sensing that produces actionable intelligence from unprocessed data and deployment capabilities at cloud and edge computing platforms [13]. Essentially, privacy is just as important in this context as the sensor data from the smartphone is. The reader surmises that by employing this thread paradigm, a rogue entity is trying to reveal the identities of contributors to critical reports..

B. Architecture of YOLO Models

The YOLO architecture as fig. 3, resembling GoogleNet, comprises 24 convolutional layers, four max-pooling layers, and two fully connected layers. Its sequential arrangement processes input images (resized to 448x448 pixels), extracting features. Employing 1x1 and 3x3 convolutional layers with ReLU activation, the model employs batch normalization and dropout techniques to enhance performance and curb overfitting, ensuring robustness and improved generalization.

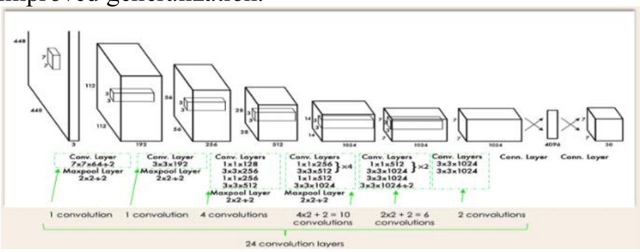

Fig 3. Architecture of YOLO model

C. Mobile Crowd Sensing for Smart Urban Mobility

Here, we have concentrated on the key areas that affect citizen mobility and city dynamics, such as traffic, air quality, and citizens' curriculum activated while on the move. This topic shows how crowdsourcing geographic information, in which residents of urban areas use their mobile, computing, communication, and sensing devices to collect geo-referenced information, can also benefit other expectations of urban life and planning. Include the Drive Sensation application as well. This tool assists in detecting dynamic traffic information and provides dynamic navigation services depending on the state of traffic at the moment. Each of these services suggests a way to stop this from happening.

D. Economical Techniques for the Participant Selection Issue in a Mobile Crowd Sensing Setting

This Algorithm topic offered some participant selection to be executed at each participant, where the players opt to be active after exchanging messages with their neighbors. However, in order to address the ILP, where mobile participants are spread out over a large area. In this case, the amount of time available may be insufficient, making the selection of activity participants nearly impossible in real life. The simulation results show that DPS is close to the optimal one, which provides some coverage area. The ILP is objective in terms of execution time as well as the need for all participants' information to be in one location.

E. Protection of Privacy in Mobile Crowd Sensing

In this section, we investigated the possible privacy concerns with MCS and determined its properties. To gain an understanding of the current survey work, we here develop a set of privacy preservation requirements in MCS. We found that in addition to privacy concerns, accountability and efficiency must be met when creating privacy preservation schemes in order to secure privacy effectively and comprehensively. To evaluate the new mandate, we surveyed MCS users about privacy protection and looked at the advantages and disadvantages of current options. We put together a summary of several issues and suggested crucial research avenues for further investigation on this topic.

F. Version of YOLO model, Why Does YOLO Outperform Object Detection Algorithm

While there may have been updates or revisions to the YOLO model since then, it's crucial to keep in mind that developments in deep learning and computer vision are still ongoing. To make sure you are as informed as possible about the YOLO model and other deep learning models, it is always a good idea to review the most recent studies and available materials. There are several YOLO models are available but, in our research, we taken YOLO V5, V8, and V8 NAS for comparison. Several factors contribute to its superior performance over other object detection algorithms. They are as follows: training strategy, flexibility, contextual information, speed, unified approach, and strong feature representation. Although there are many benefits to YOLO, it's vital to remember that other object detection algorithms, such as SSD (Single Shot Multi Box Detector) and Faster R-CNN, are superior in particular areas. Which algorithm works best for a given task will depend on its particular requirements.

- Classification-based algorithms
- Regression-based algorithm

IV. RESEARCH METHODOLOGY

A. Propose System Flow

Three most common versions of the YOLO model are YOLO V5, V8 and V8 NAS. These models are used in our study because they have made notable progress in object detection and person counting. The YOLO V5 is acknowledged for its effectiveness and superior precision in object identification, utilising cutting-edge architecture to achieve real-time performance without sacrificing quality. Our goal is to utilise its velocity and accuracy to enhance the number of individuals in gathering situations using Crowd Counting technologies. Additionally, integrating YOLO V8, which builds on earlier successes, enhances detection accuracy through advanced network architectures and optimizations. This inclusion allows exploration of its potential for more precise person counting. Our research incorporates YOLO V8 NAS, utilizing Neural Architecture Search to automate optimal neural network discovery for object detection tasks. By leveraging this model, we aim to further improve accuracy and efficiency in person counting, capitalizing on its automatic architecture search capability. Through the comprehensive use of these YOLO models, our study delves into their strengths and capabilities, aiming to identify the most suitable model or combination for enhancing the accuracy and robustness of crowd counting in meeting scenarios. This comparative analysis enables a thorough understanding of their performances, aiding in the selection of the most effective model or blend of models for precise crowd density tracking.

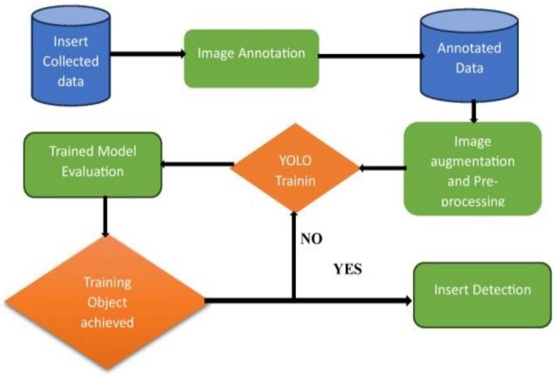

Fig 4. Data Preparation for Model Training

We compile a dataset of pictures or videos showing various meeting situations where keeping track of crowddensity requires for person counting. Preprocessing techniques including scaling, normalization and augmentation are applied to the data to improve consistency and model performance. Next, as shown in fig.4 above, we have choosen the appropriate Yolo model variant, like Yolo V5, Yolo V8 or Yolo V8 NAS, based on how well they perform in tasks involving object detection and human counting.

B. Data Collection and Preparation

We collect comprehensive video footage from densely populated areas to drive accurate person counting. Strategically placed cameras capture the dynamic crowd behavior and density, providing critical insights into patterns and movements within these bustling environments. This visual data forms the backbone of our analysis, training our algorithms to precisely detect and count individuals amidst varied crowd densities and configurations. Through rigorous analysis of this footage, we refine our methodologies, aiming to optimize our systems' accuracy in gauging crowd sizes. This data-centric approach not only enhances our technological capabilities but also underscores the importance of data-driven strategies in understanding and managing crowded spaces.

The goal of the YOLO algorithm is to predict both the object's class and the bounding box that indicates where the object is located in the input image. It uses four point values to identify each bounding box; (b_x, b_y)-Bounding box centre, b_w- Box Width and b_h-Box height. Furthermore, YOLO forecasts the probability of the prediction Pc in addition to the corresponding c number for the anticipated class.

Bounding Boxes

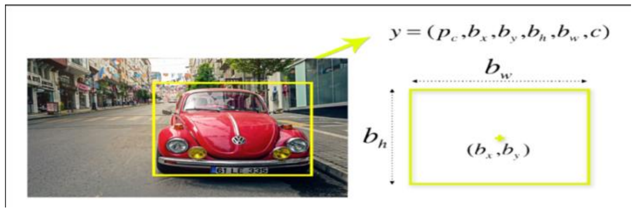

Fig 5. Bounding Boxes Parameter

For object detection, parameters b_x, b_y, b_w and b_h establish a bounding box around the "vehicle" object in the image as shown in fig.5. Including class labels like Truck, Car, or Cycle categorizes and delineates the object, enabling accurate identification within the specified classes.

1) Grid Cell Concept

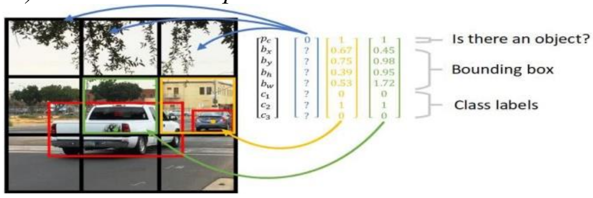

Fig 6. Bounding box parameter of grid cell

As illustrated in fig. 6, YOLO generates a 19x19 grid on the image; however, in this instance, we generate a 3x3 grid cell for comprehension. Consequently, we have separated the image into nine sections. Fig. 6's first element will be 1 if an object is present and 0 otherwise. The values bx, by, bh, and bw will come next, and the final three will indicate the class that is present in that cell according to the classes we have. The second value is 1 because we have a portion of the car identical to what is in the middle cell.

2) Anchor Boxes

Fig 7. Anchor Boxes

Cutting-edge object detection models like YOLO rely on anchor boxes as predefined bounding boxes as shown in fig. 7. These boxes filter predictions based on class scores, assist in non-maximum suppression, and refine localization through intersection-over-union (IOU) calculations. Anchors, tiled across the image according to bounding box classes, aim to identify objects with higher IOU values. Dataset division into training, testing (20%), and validation (10%) subsets aids model learning, evaluation and parameter tuning, preventing overfitting.

V. EXPERIMENTAL ANALYSIS

A. YOLO V5

YOLOV5, an advanced object detection algorithm, employs a modified CNN architecture featuring a Cross Stage Partial CSP-Darknet backbone and Spatial Pyramid Pooling (SPP) layer. [14] The architecture consists of three segments: CSP-Darknet for feature extraction, PANet in the Neck and YOLOV5 section for multi-scale object identification, and the Head responsible for final object recognition and anchor box generation with class labels.

YOLOV5 enhances upon earlier versions by adopting the EfficientDet architecture, derived from EfficientNet, enabling heightened accuracy and broader object category generalization. This complexity facilitates superior performance and wider object categorization abilities.

B. YOLO V8

The latest addition to the YOLO lineup, YOLO V8, stands out for its one-pass object prediction in images. Trained on expansive datasets like COCO and ImageNet, it excels in recognizing pre-trained classes while easily adapting to new ones. YOLOV8, with anchor-free detection and mosaic augmentation, showcases speed, compactness, and the ability to attain high accuracy, all while being accessible for single-GPU training. Maintained by Ultralytics, it's an open-source model distributed under the GNU GPL, enabling widespread sharing and modification. Additionally, the model's architecture can be visualized via ONNX, offering a universal representation for machine learning models.

C. YOLO V8 NAS

YOLO V8 NAS combines Neural Architecture Search (NAS) with YOLO for automated neural network design. This method explores optimal architectures by tweaking layer numbers, configurations, and other parameters to enhance object detection performance. By automating architecture design, it tailors YOLO models to specific tasks or datasets, improving efficiency and accuracy in object detection through customized architecture optimization.

D. EVALUATION METRICS

We evaluated the performance of object detection or recognition models using evaluation metrics[15] such as MAP (mean average precision), precision, and recall in order to compare the models

- .mAP(mean Average Precision), assesses an object detection model's overall accuracy by averaging precision scores across different categories.
- Precision evaluates the correct identification of positive cases among total predicted positives, emphasizing model correctness.
- Recall measures the model's capability to detect all positive instances, depicting the ratio of correctly identified positives among total actual positives. Greater precision signifies fewer false positives, while higher recall implies fewer false negatives.

The mAP metric consolidates precision-recall trade-offs for each class, offering a comprehensive evaluation of the model's performance[17] across diverse categories in object detection.

E. PERFORMANCE EVALUATION

Table-I's results show that the YOLO V8 NAS is outperforming other models, as seen by the graphical representation in fig. 8.

TABLE-I. YOLO V5, V8, V8 NAS SIMULATION EXPERIMENTS

Algorithm	Epoch	Batch	mAP .5	mAP (.5-.95)
YOLO V5	12	34	82.6%	68.02%
YOLO V8	12	34	89.28%	70.31%
YOLO V8 NAS	12	34	95.7%	92.4%

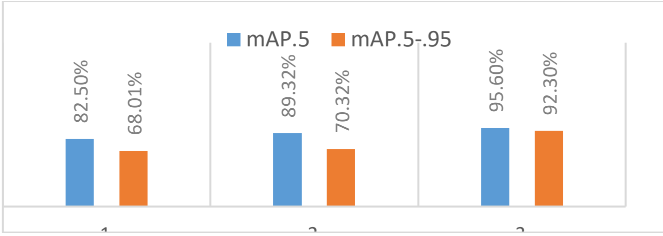

Fig 8. mAP value of YOLO V5, V8, V8 NAS

Table II displays the mAP , Precision and Recall values. A graphic comparison of these three models is provided in fig. 9, which is provided below.

TABLE –II. VALUES OF mAP, PRECISION AND RECALL AFTER MODEL SIMULATION

Models	Epoch	Batch	mAP (%)	Precision	Recall
YOLO V5	12	34	82.51	73.11%	76.8%
YOLO V8	12	34	89.33	88.8%	63.8%
YOLO V8 NAS	12	34	95.6	93.2%	85.5%

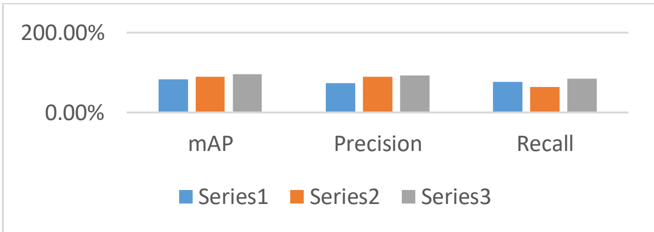

Fig 9. mAP, Precission, Recall value of YOLO V5, V8, V8 NAS

After successfully implemented our proposed model, it shows outstanding performance in counting number[18-20] of person present in crowed area which are shown in below fig. 10.

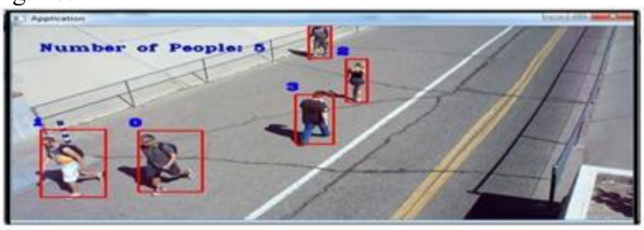

Fig 10. Result after Model Implementation

We motivated from Balakrishnan et al. [16] where the authors proposed a model YOLO V8 which taken the realtime data using sensors from crowded are which gives mAp value of 89.28% as compare his research our model i.e., YLO V8 NAS which take data from sensing device (IoT) and simulate and gives mAP value of 95.7%. which is more than previous research.

VI. Conclusion and Future work

This study highlights the effective collaboration between IoT-driven data gathering and advanced YOLO-based object detection for real-time crowd monitoring. The assessment of YOLO versions V5, V8 and V8 NAS reveals YOLO V8 NAS's exceptional 95.6% mAP, attributed to its NAS-optimized architecture for precise crowd counting. Future work aims at creating a software-hardware model which will be a python-based software and analyze sensor data for crowd density, integrating with IoT platforms. Simultaneously, hardware using IP Camera sensors will capture real-world data. This innovative fusion will show promise across industries like transportation, marketing, and smart city management, promising improved crowd monitoring and safety.

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